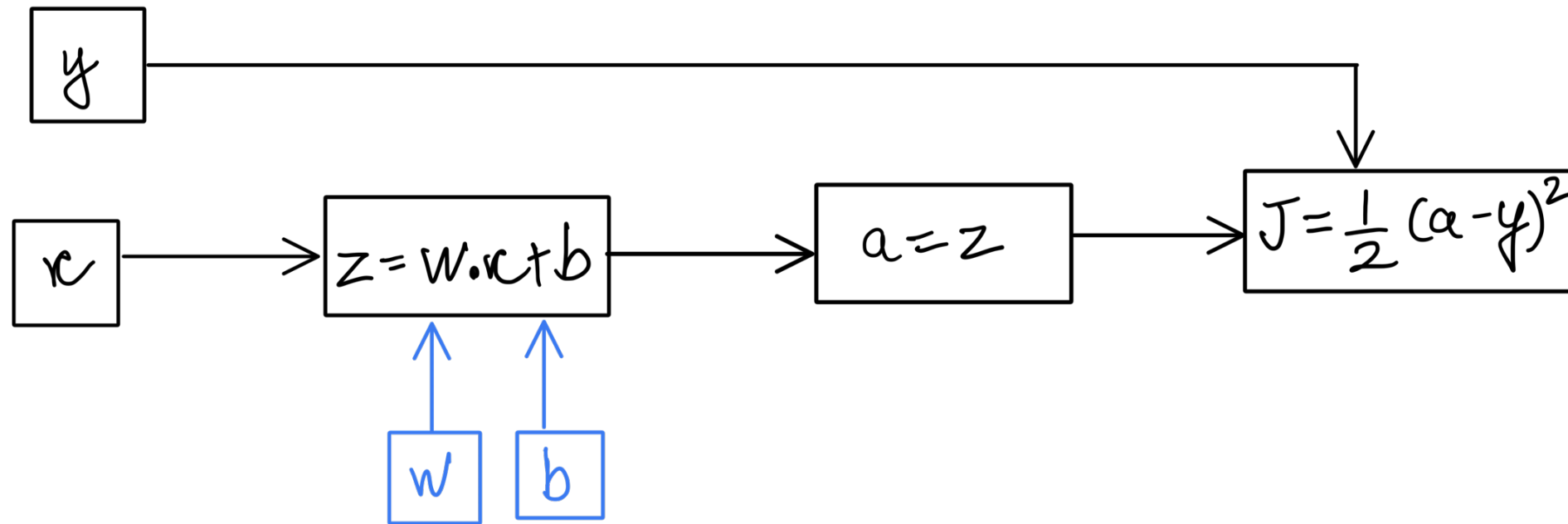


★ Computational Graph



when $x = -2$, $y = 2$, $w = 2$ & $b = 8$

$$\Rightarrow z = 4, J = 2$$

For small change ϵ in a , J becomes $\frac{1}{2} (2 - \epsilon)^2$

$$\frac{\delta J}{\delta a} = \frac{1}{2} (2 - \epsilon)^2 - \frac{1}{2} (2)^2 = \frac{1}{2} (\epsilon^2 - 4\epsilon)$$